# STALKS OF A QUASI-COHERENT MODULE COMPUTED BY TENSORING THE STALK ON AN ETALE NEIGHBOURHOOD

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*Editorial note.* All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

*Editorial note.* Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

*Source reference.* Original lemma 04KF: sources/stacks-chow/tags.txt, source lines 5971–5971. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original definition 04JY: sources/stacks-chow/tags.txt, source lines 5954–5954. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original lemma 03LZ: sources/stacks-chow/tags.txt, source lines 4731–4731. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *Let $S$ be a scheme. Let $X$ be an algebraic space over $S$. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Let $x \in |X|$ be a point and let $\overline{x}$ be a geometric point lying over $x$. Finally, let $\varphi : (U, \overline{u}) \to (X, \overline{x})$ be an étale neighbourhood where $U$ is a scheme. Then*

$$(\varphi^*\mathcal{F})_u \otimes_{\mathcal{O}_{U,u}} \mathcal{O}_{X,\overline{x}} = \mathcal{F}_{\overline{x}}$$

*where $u \in U$ is the image of $\overline{u}$.*

*Proof.* Note that $\mathcal{O}_{X,\overline{x}} = \mathcal{O}^{sh}_{U,u}$ by Lemma 04KF hence the tensor product makes sense. Moreover, from Definition 04JY it is clear that

$$\mathcal{F}_{\overline{u}} = \operatorname{colim}(\varphi^*\mathcal{F})_u$$

where the colimit is over $\varphi : (U, \overline{u}) \to (X, \overline{x})$ as in the lemma. Hence there is a canonical map from left to right in the statement of the lemma. We have a similar colimit description for $\mathcal{O}_{X,\overline{x}}$ and by Lemma 03LZ we have

$$((\varphi')^*\mathcal{F})_{u'} = (\varphi^*\mathcal{F})_u \otimes_{\mathcal{O}_{U,u}} \mathcal{O}_{U',u'}$$

whenever $(U', \overline{u}') \to (U, \overline{u})$ is a morphism of étale neighbourhoods. To complete the proof we use that $\otimes$ commutes with colimits. □

*Date*: stacks-project:spaces-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.